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Natural Form(s): case study of a spatial framework composed of naturally grown forked branches

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Abstract

Wood as a construction material not only serves as a sustainable resource but also offers the potential to develop complex structural models (and architectural spaces) that are derived from its complex material logic. This paper proposes an experimental methodology for the design of spatial frameworks that utilize natural forked branches as structural components. Computational methods such as 3D-scanning and parametric structural analysis tools allow designers to cope with the irregular complexity of the natural material. This paper presents a comprehensive digital workflow leading to the robotic fabrication of a medium scaled demonstrator.

Keywords: Naturally grown form, complex wood structures, spatial frameworks, discrete element aggregation, 3D Scanning, parametric engineering, structural optimization, digital fabrication

1. Introduction

The structural system of trees responds to multiple forces forming a complex configuration of structural members, resulting in an optimized form, with forking branches acting as optimized joints (Mattheck [1]) (Müller *et al.* [2]).

Based on concepts previously presented (Allner and Kroehnert [3]), this paper expands the research with a design and fabrication process of an experimental spatial framework formed by tangentially joined branch parts. A digitized set of hornbeam branches is utilized as components in a synthetic formation. This is a project in the tradition of architectural research attempting to utilize wood in its natural form (Mollica and Self [4]). It also explores how design control becomes relative to the constraints introduced by the available inventory, determining a dynamic design process in which constraints and design goals are negotiated in a feedback system.

Adapting the structural logic of natural trees to architectural applications, originally cantilevering branches are reassembled to form closed loops constituting a cellular framework with improved structural integrity and general bracing. For the purpose of this paper, the demonstrator represents a case study of a regular lattice of tetrahedron cells.

2. Digitization and Cataloging

The workflow starts with a collection of 3d scan models of the physical branch parts, which have been prepared by flush cutting the ends and with marking of several reference points.

In an automated process centerline models (Figure 1) are generated from the mesh geometries: Centroids of co-parallel section outlines define vertices of polylines. For an optimal matching of the indi-

vidual branches to their averaged placeholders within the spatial lattice grid according to the structural load scenario each branch arm's diameter is generated and represented by a circle.

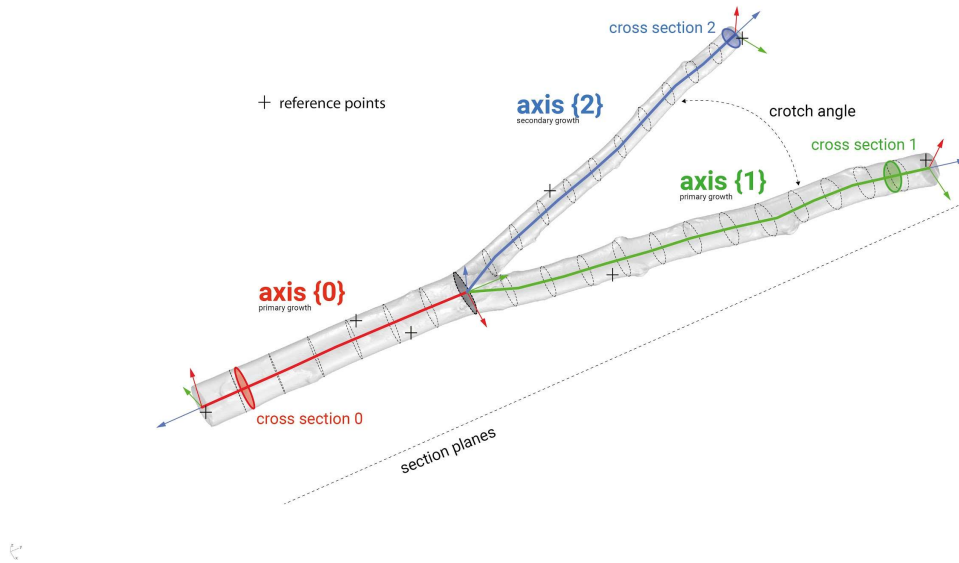


Figure 1: Automated generation of centerlines and additional necessary data

In the digital model points are defined at the markings registered in the scan texture.

These serve as geometric reference for matching the digital and the analog models at any step of the process and allow to monitor changes in shape caused by varying humidity and drying. With this method a high precision placement during CAM processing is achieved despite the complex irregular form.

The digitizing process returns a catalog containing a large number of unique elements, each with a similar topology, but different dimensions, angles, cross-sections and orientations of the individual branches (Figure 2).

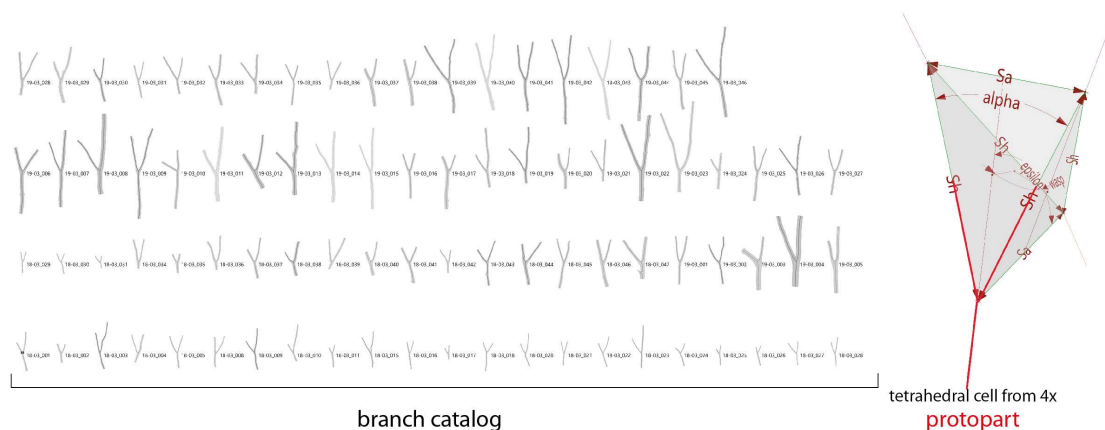


Figure 2: branch catalog and simplified parts assembled into tetrahedral cell

3. Aggregation and Structural Evaluation

As an intermediate step to arrive at a final formation, an initial aggregation of identical typical elements or proto-parts is generated. The proto-parts are used as place holders for the actual, unique specimen to be positioned in a second step.

All collected branches for this case study share the same anatomic principle of a single bifurcation. This typology can be approximated into a Y-shaped axis model with three connected line segments. Four of these elements are assembled into one tetrahedral cell which is aggregated into potentially endlessly structures (Figure 3). Amongst many possible formations composed of branch parts the presented investigations focus on this one specific instance of a topologically repetitive/regular lattice of tetrahedral cells as a case study.

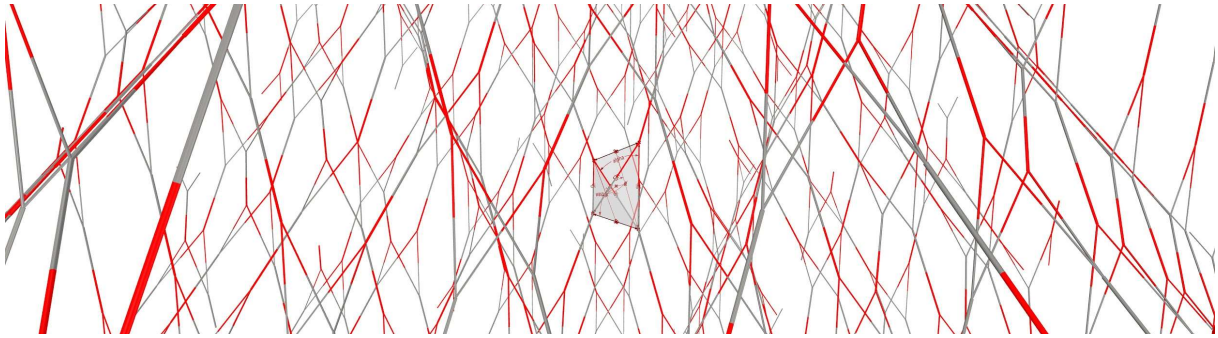


Figure 3: spatial lattice aggregation from proto-part tetrahedral cells

For the aggregation of simplified elements the Wasp plug-in for Grasshopper is used. This tool for discrete design was developed by Rossi and Tessmann [5].

One averaged Y-shaped proto-part is constructed based on the average crotch angle and the longest branch arm length retrieved from the set of scanned elements. An aggregation of those identical proto-parts approximates the extents and global density of the final formation.

Parts of the structure which have only one connection with another single branch are eliminated for redundancy. This ensures that there are no “flying” elements in the aggregation weakening the overall structural performance.

The global proto-part aggregation is then evaluated in Karamba3D for Grasshopper, a plugin developed by Preisinger *et al.* [6]. Different load case scenarios are evaluated (Figure 4). Generic assumptions of gravity and horizontal wind loads (a) are applied along with various vertical load-cases such as a “roof” dead loads (a), “floor-plate” live loads (b) and randomly distributed live loads throughout the structure (c). These calculations are done applying the average cross section to all elements.

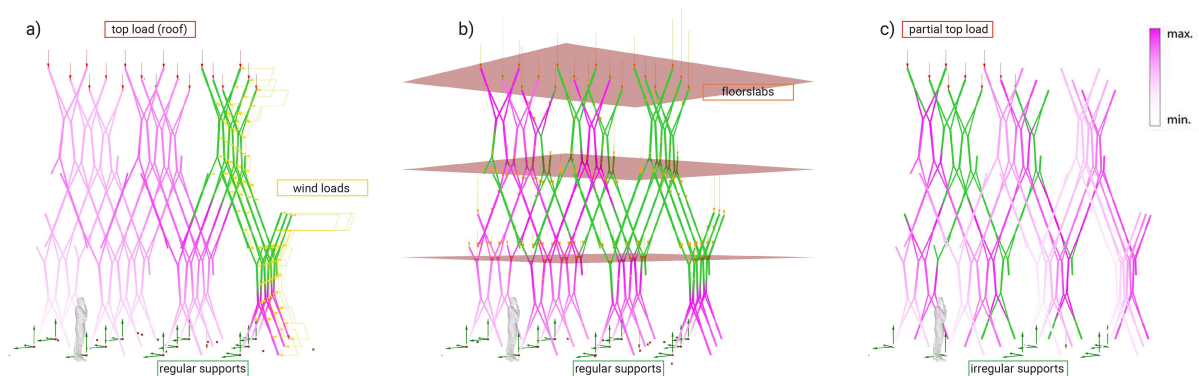


Figure 4: different load case calculations with Karamba3D - displacement visualization

3.1. Part Replacement

Subsequently, each instance of proto-parts is replaced by one of the real specimen from the catalog. Local utilization as shown by the simplified structural analysis is used as defining parameter for the final combination of available elements. The analysis affects a layout of required cross section distribution (Figure 5). Each member in the set is then positioned according to the best fit average cross section per part.

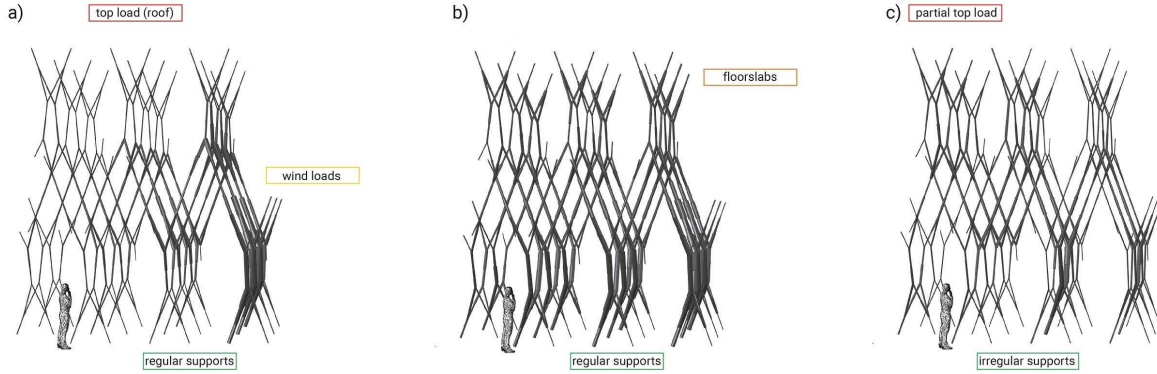


Figure 5: initial cross section requirements for load case scenarios

3.2. Gap Compensation with Dynamic Relaxation Method

Replacing idealized modular elements with actual specimen, each with a unique form, generates discontinuities in the framework. To apply the dynamic relaxation method the Grasshopper plug-in for Kangaroo (Senatore and Piker [7]) is used. Gaps and discontinuities are compensated by translating the aggregation topology into a constraint-based model in Kangaroo. Parts are defined as rigid bodies with springs connecting end segments. In the iterative simulation the springs compress, closing gaps and forcing ends to overlap. Additionally a constraint is set for overlapping end segments to align (Figure 6).

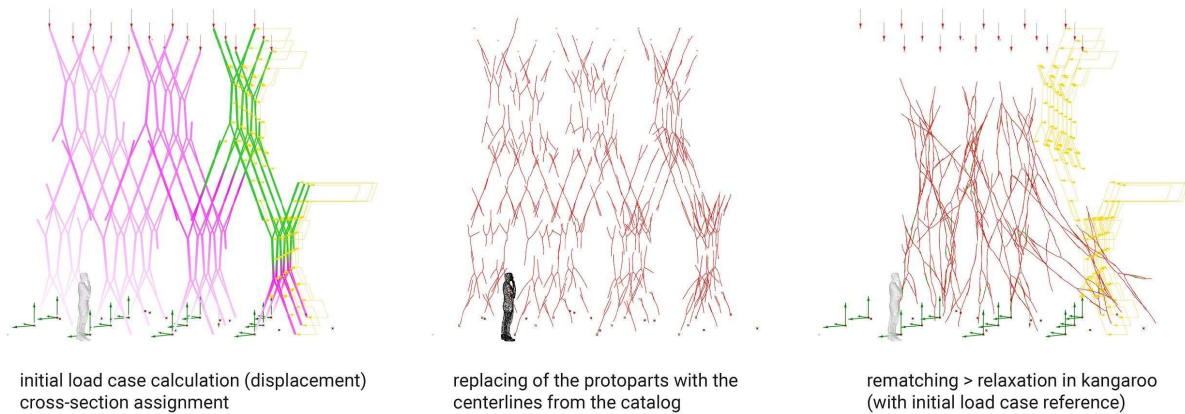


Figure 6: load case calculation, replacement of proto-parts and realignment of the real branch-centerlines

After relaxation the 3D scanned mesh models are mapped to the locations of their respective axis model.

4. Results

As expected the deformed model shows a significant transformation of the initial regular aggregation. Dependent on the scenario large variations of the initial form occur, representing a combinatory adaptation of the same set to the respective force conditions (Figure 7).

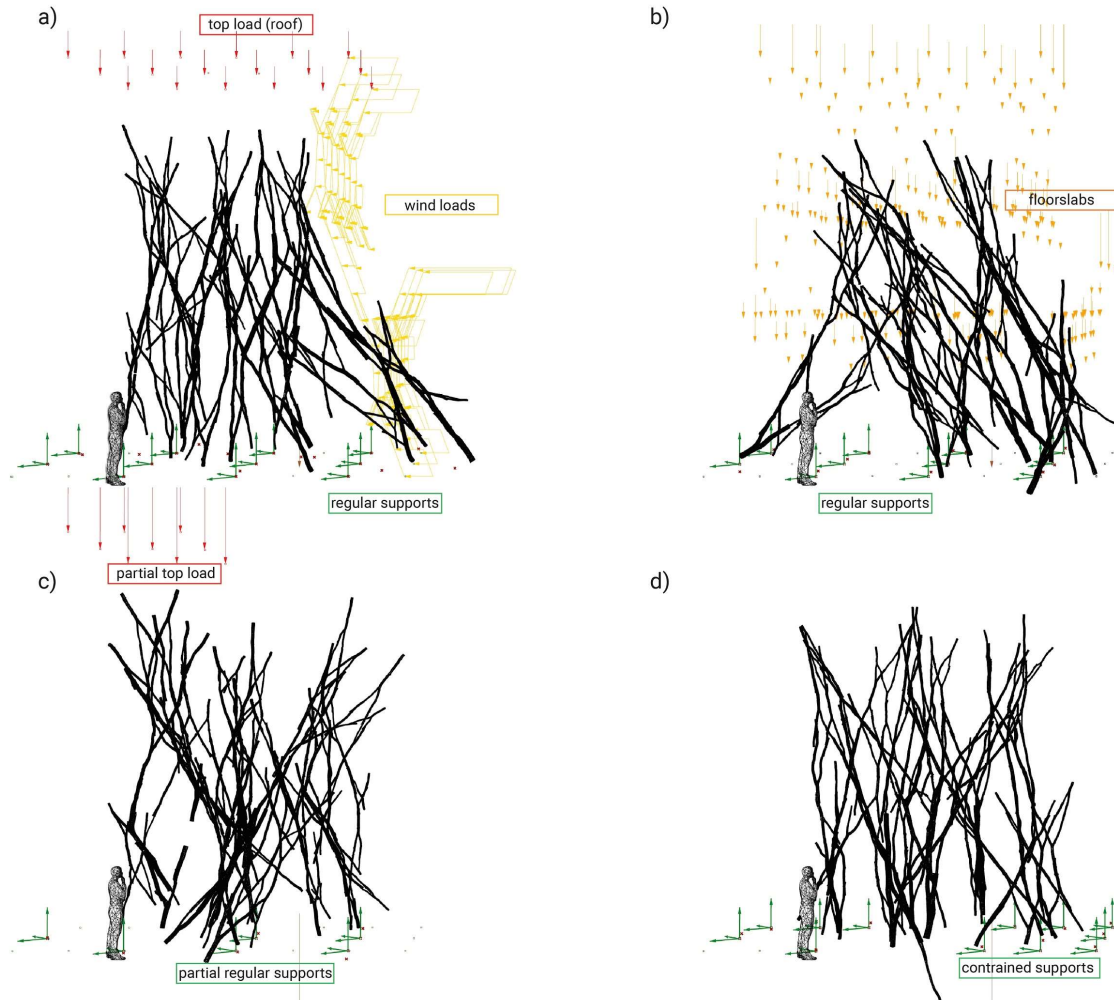


Figure 7: replaced and realigned branch-set from the catalog according to different load case scenarios

For the same load case scenario the required cross sections are visualized as circular profiles and overlaid with the actual branch geometry. The comparison shows that in the calculated load case scenarios the necessary cross sections are much smaller than the available branches. (Figure 8) There are two possible explanations for this that need further evaluation. The final structure is smaller than the initially calculated model as the maximum branch arm lengths are taken for the proto-part definition. Therefore the structure is globally more compact and thus more rigid against forces. Another hypothesis is that the structure is more stable because of its irregularity compared to a regular structure.

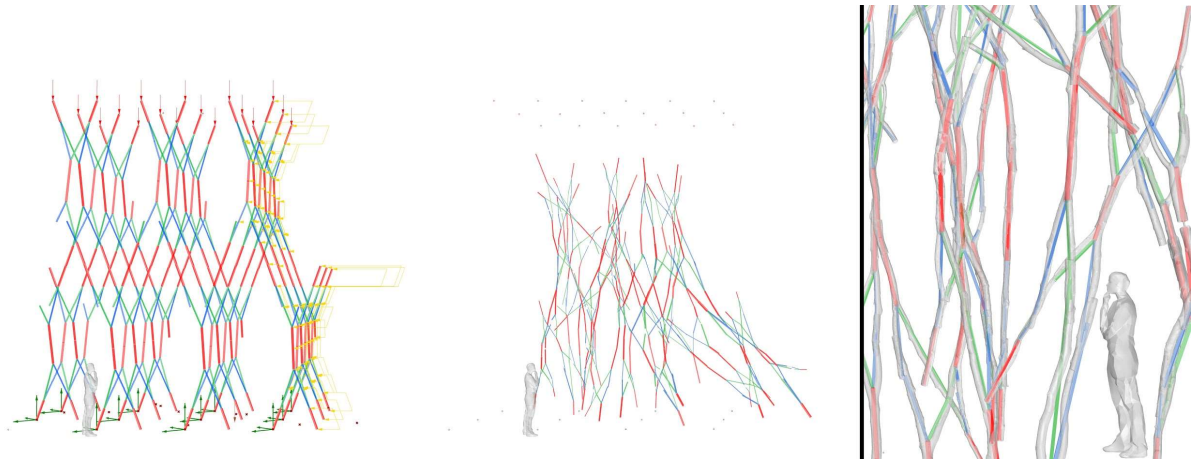


Figure 8: comparison of cross sections in the regular (left) and in the final aggregation (middle). Required cross sections in color overlaid with actual branch geometries

As an optimization of the found formation each branch element is evaluated against its orientation in the framework. Each branch element has a different cross section at index 1 and 2 (see Figure 1), ideally the larger cross section should be assigned to the larger utilization value. Instances with a mismatching association are flipped. Then relaxation is performed again on the modified aggregation with flipped elements. This process is repeated in a loop expected to render an optimized aggregation after a number of iterations. However the change in global geometry caused by the flipping is too drastic to produce meaningful improvements.

With an inventory restricted to elements with identical branch segment lengths the geometric change in the aggregated form could be minimized. A possibly more effective approach would be to reconsider the choice of cross sections as the only selection criteria for the positioning of elements. A multi criteria assessment also including geometric features of branches would increase the level of control for structural optimization.

5. Conclusion and Outlook

At the moment of writing this paper a workflow and techniques for CNC and robotic fabrication of a medium scale demonstrator composed of approximately 30 parts are developed (Figure 9).



Figure 9: fabrication tests with a 3 axis milling machine (left) and a robotic arm (right)

Currently improvements and additional functionality of the aggregation algorithm (Wasp) are under development to integrate more diverse categorization of the set as well as goal based iterative aggregation principles.

The material properties of a common Hornbeam structural element, was assigned for the structural analyses calculations. Due to the peculiarities in such an untreated and non-standard material, a further material analysis of the found branches is important to gauge its actual structural performance.

Another aspect to consider is the difference of stem and branch wood on a cellular level as indicated in the research of Lichtenegger *et al.* [8]. Whereas stem wood is naturally optimized for stiffness and strength, branch wood is optimized for extensibility and bending. Therefore further investigations into the necessary parameters for a valid calculation have to be undertaken.

The preliminary results show the potential of the developed methodology for the exploration of structural design with irregular parts. The proposed workflow could be expanded to other geometries in the context of inventory constraint design methods for reused structural materials as introduced by Bukauskas *et al.* [9]

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References

- [1] C. Mattheck, *Design in Nature, Learning from Trees*, Springer, Berlin, 1998
- [2] U. Müller, W. Gindl-Altmutter, J. Konnerth, D. Kaserer, J. Keckes, “*Stamm-Astanzbindung – eine biologisch optimierte Struktur mit hoher mechanischer Leistungsfähigkeit*“, 10.GraHFT’14, 2014.
- [3] L. Allner and D. Kroehnert, “Conceptual Joining: Branch Formations,” in *Creativity in Structural Design: Proceedings of the IASS Annual Symposium 2018*, Boston, USA
- [4] Z. Mollica and M. Self, “*Tree Fork Truss*” in *Advances in Architectural Geometry*, S. Adriaenssens, F. Gramazio, M. Kohler, A. Menges, M. Pauly (eds.), VDF, 2016
- [5] A. Rossi and O. Tessmann, “*From Voxels to Parts: Hierarchical Discrete Modeling for Design and Assembly*.”, *International Conference on Geometry and Graphics* (pp. 1001-1012). Springer, Cham., 2018.
- [6] C. Preisinger, „*Linking Structure and parametric Geometry*“, *Architectural Design*, 83: 110-113, doi: 10.1002/ad.1564, 2013.
- [7] G. Senatore and D. Piker, “*Interactive real-time physics: an intuitive approach to form-finding and structural analysis for design and education*,” *Computer-Aided Design*, vol. 61, 2015
- [8] H. Lichtenegger, A. Reiterer, S.E. Stanzl-Tschegg, and P. Fratzl. *Variation of Cellulose Microfibril Angles in Softwoods and Hardwoods-A Possible Strategy of Mechanical Optimization*, *Journal of Structural Biology* 128 (3), 257-269 (1999)
- [9] A. Bukauskas, P. Shepherd, P. Walker, B. Sharma, J. Bregulla, “*Inventory-Constrained Structural Design: New Objectives and Optimization Techniques*” *Proceedings of the IASS Symposium 2018 Creativity in Structural Design*, 2018